Supplementary Material Concerning the Paper

# Synthesis of Optimal Stealthy Sensor Attackers Against Opacity with Asymmetric Observability

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## I. Proof of Proposition 1

1) First, we consider $y_e = y_0$, i.e., $\pi = \varepsilon$ and $\xi_A = UR_A(\{x_0\}, z_0)$. Clearly, $\pi|_A = \varepsilon$. By Definition 3,

$$C_A(\varepsilon) = \left\{ x \in X \mid \begin{array}{l} \exists \pi' \in L_e(S/G, \Lambda) \\ \text{s.t. } \pi'|_A = \varepsilon \wedge \delta(x_0, \pi'|_G) = x \end{array} \right\}.$$

Furthermore, due to Assumption A4, we have

$$C_A(\varepsilon) = \left\{ x \in X \mid \begin{array}{l} \pi'|_G \in ((\Sigma_{uo}^A \cap E_S(z_0))^* \\ \wedge \delta(x_0, \pi'|_G) = x \end{array} \right\}.$$

Trivially, $C_A(\varepsilon) = UR_A(\{x_0\}, z_0)$. Thus, $\xi_A = C_A(\pi|_A)$.

Next, we consider $y_e$ that is reached from an environment state $y'_e = (\xi'_S, \xi'_A, x', z')$ via a path string $\sigma_n \widetilde{\sigma'_n} \in \Sigma\tilde{\Sigma}$ and $y'_e$ is reached from $y_0$ via a path string $\pi_1 \in (\Sigma\tilde{\Sigma})^*$ with $\xi'_A = C_A(\pi_1|_A)$. In other words, $\pi = \pi_1 \sigma_n \widetilde{\sigma'_n}$. There are two cases to be analyzed: 1)$\sigma_n \notin \Sigma_o^A$; and 2) $\sigma_n \in \Sigma_o^A$. In the case of $\sigma_n \notin \Sigma_o^A$, we can see that $\xi_A = \xi'_A$ and $\pi|_A = \pi_1|_A$. Since $\xi'_A = C_A(\pi_1|_A)$, it holds that $\xi_A = C_A(\pi|_A)$. Now, we consider the case of $\sigma_n \in \Sigma_o^A$. It is $\xi_A = UR_A\left(NX_{\sigma_n}(\xi'_A), z\right)$ by Definition 6. Let $\omega = \pi|_A$ and $\omega_1 = \pi_1|_A$. Clearly, $\omega = \omega_1 \sigma_n \widetilde{\sigma'_n}$. According to Definition 3,

$$C_A(\omega) = \left\{ x \in X \mid \begin{array}{l} \exists \pi' \in L_e(S/G, \Lambda) \\ \text{s.t. } \pi'|_A = \omega \wedge \delta(x_0, \pi'|_G) = x \end{array} \right\}.$$

Due to Assumption A4, we have

$$C_A(\omega) = \left\{ x \in X \mid \begin{array}{r} \exists x' \in C_A(\omega_1), \exists \pi'' \in ((\Sigma_{uo}^A \cap E_S(z))^*, \\ \text{s.t. } \delta(x', \sigma_n \pi'') = x \end{array} \right\}.$$

Since $\xi'_A = C_A(\omega_1)$, it is trivial to see that $C_A(\omega) = UR_A\left(NX_{\sigma_n}(\xi'_A), z\right)$. Thus, $C_A(\pi|_A) = \xi_A$.

Consequently, based on the above analysis, we conclude that $\xi_A = C_A(\pi|_A)$.

2) Similar to the proof in 1).

3) ($\Rightarrow$) Since $\xi_S = AX$, we have $y_e \neq y_0$. Thus, without loss of generality, there exists a path $y_0 \xrightarrow{\pi_1} y'_e \xrightarrow{\sigma_n \widetilde{\sigma'_n}} y_e$, where $y'_e \in Y_e$, $\pi_1 \in (\Sigma\tilde{\Sigma})^*$, and $\sigma_n \widetilde{\sigma'_n} \in \Sigma\tilde{\Sigma}$. Clearly, $\pi = \pi_1 \sigma_n \widetilde{\sigma'_n}$. Let $y'_e = (\xi'_S, \xi'_A, x', z')$. According to Definition 6, we have $(z = z_{\text{det}}) \vee (\sigma'_n \in \Sigma_o^S \wedge NX_{\sigma'_n}(\xi'_s) = \emptyset)$.

a) Consider $z = z_{\text{det}}$. By Definition 6, it indicates that $\sigma'_n \in \Sigma_o^S$ and $\delta_S(z', \sigma'_n)$ is not defined. Moreover, it is $\delta_S(z_0, \pi_1|_S) = z'$. Thus, $\pi|_S = \pi_1|_S \sigma'_n \notin P_S(L(S/G))$.

b) Consider $\sigma'_n \in \Sigma_o^S \wedge NX_{\sigma'_n}(\xi'_s) = \emptyset$. We can see that $\xi'_S \neq AX$ since otherwise $y'_e$ is a sink node in ACAG. Thus, we have $\xi'_S = C_S(\pi_1|_S)$. Furthermore, we have $NX_{\sigma'_n}(C_S(\pi_1|_S)) = \emptyset$. This indicates that $\forall \alpha \in L(S/G)$ with $P_S(\alpha) = \pi_1|_S$, $\delta(x_0, \alpha\sigma'_n)$ is not defined. Trivially, $\pi|_S = \pi_1|_S \sigma'_n \notin P_S(L(S/G))$.

($\Leftarrow$) By contradiction, suppose that $\xi_S \neq AX$. Thus, we have $\xi_S = C_S(\pi|_S)$. According to the proof in 2), it is trivial to see that $C_S(\pi|_S) \neq \emptyset$. It implies that $\exists \alpha \in L(S/G)$, s.t. $P_S(\alpha) = \pi|_S$, which contradicts that $\pi|_S \notin P_S(L(S/G))$. Therefore, it holds that $\xi_S = AX$.

## II. Complete ACAG and AO-ACAG for the Running Example

In this supplementary material, we present the complete ACAG and AO-ACAG constructed for the running example in our paper, shown in Fig. 1 and Fig. 2, respectively. These graphs are automatically generated by our developed tool available in [32]. Note that, for the sake of implementation convenience, we write the state estimates $\xi_{Si}$ and $\xi_{Ai}$ as $S_i$ and $A_i$, respectively, and their corresponding sets are detailed in Table I. We observe that $S_i$ in Table I is a set of state pairs, which differs from its definition. These variations are adopted to facilitate the implementation while preserving the core logic of the proposed algorithms. Specifically, each pair in $S_i$ consists of a supervisor state $z$ and a plant state $x$. While the collection of all plant states (the second entries of the pairs) constitutes the supervisor estimate as defined, $S_i$ explicitly associates each supervisor state with possible plant states. Consequently, $S_0$ is the same as $S_3$ and $S_2$ is the same as $S_4$ in Table I according to the formal definition of $S_i$ in our paper. We note that, despite these additional details, the generated ACAG remains fully consistent with its definition.

TABLE I: State estimates of the attacker and the supervisor

| **Supervisor Estimates** | **Attacker Estimates** |
|---|---|
| $S_0 : \{(0, \underline{5}), (0, \underline{0})\}$ | $A_0 : \{3\}$ |
| $S_1 : \{(0, \underline{0})\}$ | $A_1 : \{5\}$ |
| $S_2 : \{(0, \underline{2}), (0, \underline{1})\}$ | $A_2 : \{6\}$ |
| $S_3 : \{(1, \underline{5}), (1, \underline{0})\}$ | $A_3 : \{8\}$ |
| $S_4 : \{(1, \underline{1}), (1, \underline{2})\}$ | $A_4 : \{4, 6\}$ |
| $S_5 : \{(1, \underline{6}), (1, \underline{7}), (1, \underline{3}), (1, \underline{4})\}$ | $A_5 : \{1, 2, 5\}$ |
| $S_6 : \{(1, \underline{6}), (1, \underline{7}), (1, \underline{4})\}$ | $A_6 : \{4, 6, 7\}$ |
| $S_7 : \{(1, \underline{7}), (1, \underline{4})\}$ | $A_7 : \{0, 1, 2, 5\}$ |
| $S_8 : \{(2, \underline{3}), (2, \underline{6})\}$ | - |

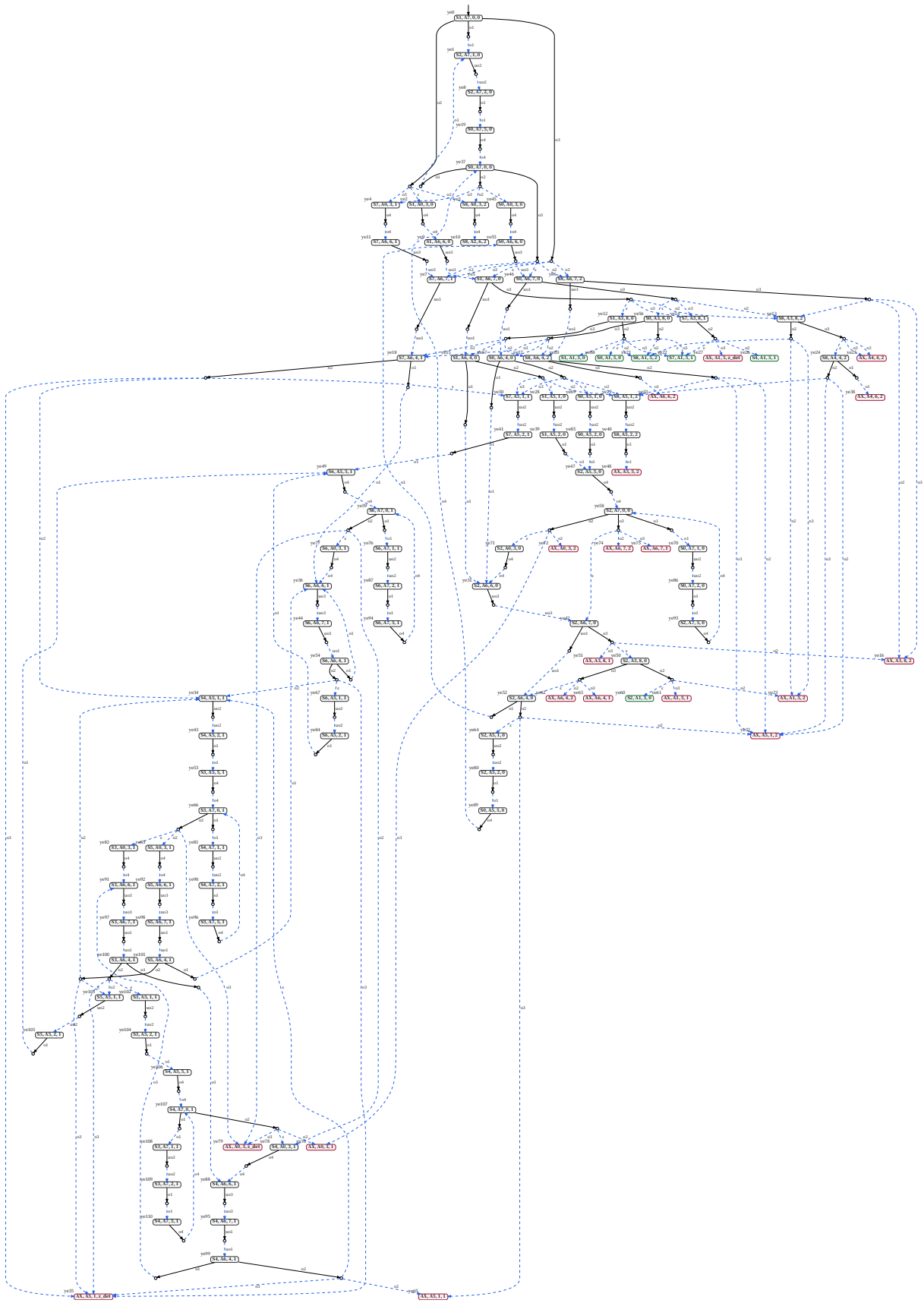

Fig. 1: ACAG $\mathcal{Y}$

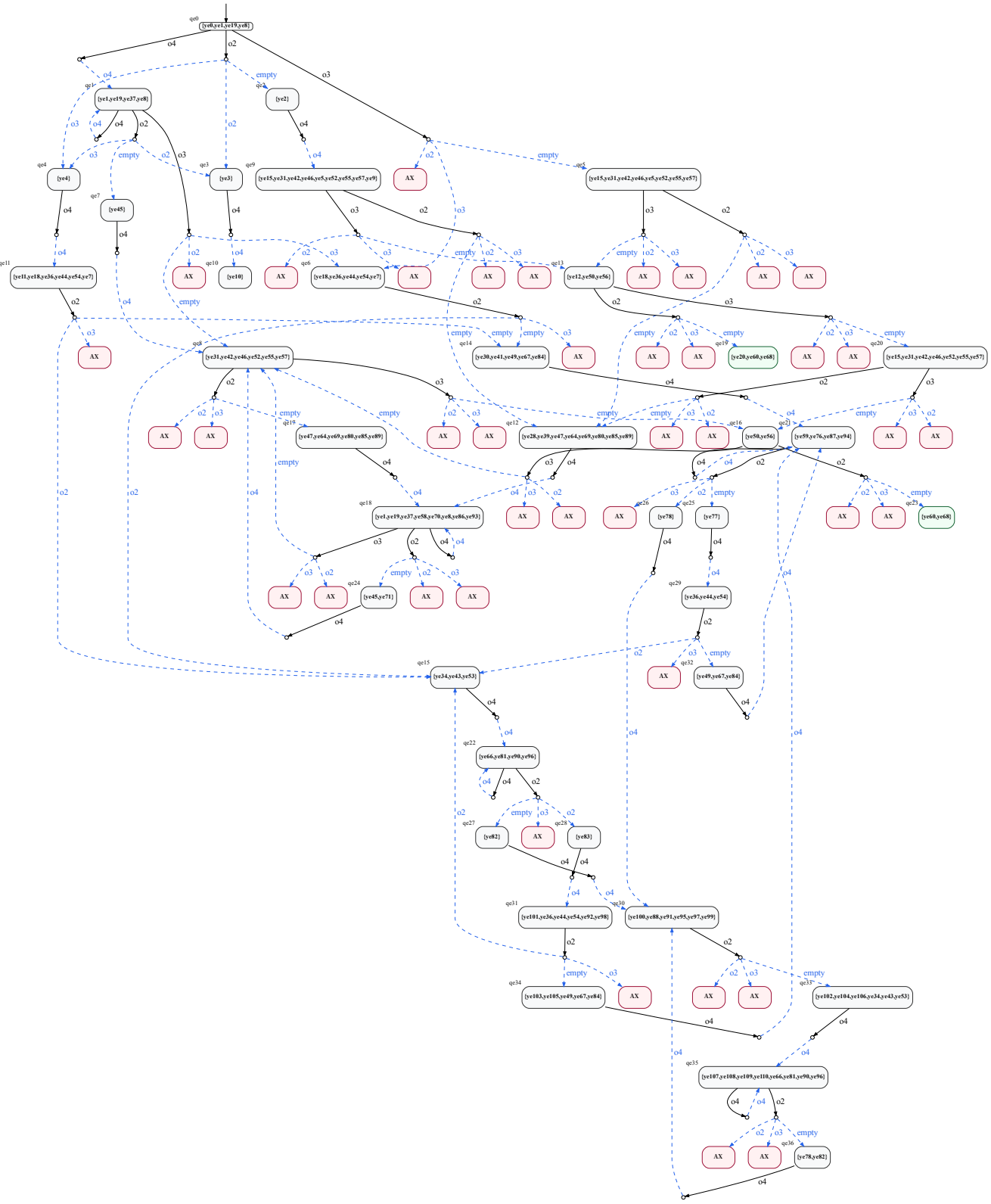


Fig. 2: AO-ACAG $\mathcal{Y}^Q$